SIEG GABRIEL GRANTOZA SOLA

sieggabrielsola12@gmail.com | www.linkedin.com/in/sieg-gabriel-sola | https://solasieggabriel.github.io/portfolio/

PROFILE

Cum Laude BS Mathematics and DOST-SEI Scholar graduate with hands-on experience building hybrid variational quantum machine learning architectures and statistical workflows in Python and R. Seeking to leverage statistics and data analysis skills in industry-leading technology and research roles.

EDUCATION

Southern Luzon State University — **Graduated: June 2026**

Bachelor of Science in Mathematics

- GWA: ~1.44 *Cum Laude*
- DOST-SEI scholar

EXPERIENCES

DOST - Advanced Science and Technology Institute — **March 2026 - April 2026**

Quantum Image Representation Intern

- Developed and benchmarked quantum encoding strategies in Python for supervised hybrid quantum machine learning algorithms, co-authoring a comprehensive study on models' performances.
- Applied statistical validation in Python and R to analyze stochastic models' outputs using non-parametric inference, post-hoc analysis, and correlation measures to assess group differences and interaction effects.

LGU - Rizal, Laguna — **June 2025 - August 2025**

Accounting Intern

- Verified financial records by cross-checking expense reports, receipts, and payroll computations against official documents while coordinating with municipal departments to resolve discrepancies and obtain approvals.
- Digitized 11 Barangays' financial data from January to June 2025 into Microsoft Excel, creating a searchable database for the Municipal Accounting Office.

Photomath Inc. — **August 2020 - January 2024**

Freelance Math Expert

- Authored thousands of step-by-step solutions for algebraic and geometric problems for a global student user base using LaTeX syntax.
- Performed quality assurance on thousands of peer-submitted solutions, ensuring strict mathematical accuracy and formatting standards.

ACADEMIC PROJECTS

Undergraduate Thesis Project — **April 2026**

Evaluating 14-Day Ahead Weather Predictions in the City of Manila: A Comparative Multivariate Study of ARIMAX and Lagged Multiple Linear Regression

- Used Python to apply PCA on 12 exogenous meteorological features to reduce dimensionality and establish 7 uncorrelated principal components for model input.
- Developed two predictive models in Python for weather forecasting, ARIMAX and Lagged Multiple Linear Regression, generated 14-day forecast for three labels, and evaluated the models using RMSE, MAE, Paired Hotelling's, and Diebold-Mariano Test, proving the superiority of the ARIMAX architecture in dampening error accumulation against high-frequency climate shocks.

TECHNICAL SKILLS

- **Mathematics**: Linear Algebra, Probability, Statistics (Parametric/Non-parametric inference, Post-hoc analysis)
- **Quantum Computing**: Qiskit, Variational Quantum Classifier, Quantum Image Encoding
- **Programming/Tools**: Python, R, Microsoft Tools (Excel, Power BI), LaTeX (Overleaf, Texmaker), Basic SQL querying

CERTIFICATIONS and PROFESSIONAL DEVELOPMENT

Civil Service Professional Eligibility (94.60%) — **Passed: May 2025**

Civil Service Commission